

[Tree-like Causal Latency Model (TCLM) and Dual-Pulse Synchronization Protocol (DPSP)] From Causal Latency Deconstruction to Intergalactic Deep-Space Chronometry Engineering

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Manifesto

Newton was right, and Einstein was also right.

The Causal Tree is a framework that allows both to hold true simultaneously.

Abstract

This paper aims to establish a complete spacetime causal reconstruction theory and its accompanying deep-space chronometry engineering standard. The paper points out that the deduction in Special Relativity stating "superluminal motion ($v > c$) leads to time reversal" is essentially a misplacement of observational signal propagation latency as a physical time reversal, causing numerical domain overflow (imaginary numbers and negative time) in the Lorentz transformation under extreme conditions.

This paper proposes the **Tree-like Causal Latency Model (TCLM)**, which strictly decouples time into "topological causal order (topological trunk)" and "geometric flow rate differential (geometric branches)." The topological trunk (t_{trunk}) is not a Newtonian global absolute clock, but a strict unidirectional topological causal order operator ($\text{sgn}(\mathcal{C}) \equiv +1$) describing the bottom-level event settlement of the universe. The geometric branches represent the relative proper time rates (τ) and geometric observation latencies of local physical entities under specific gravitational potentials and velocity fields.

To advance this theory into engineering practice, this paper simultaneously proposes a top-layer communication standard: the **Dual-Pulse Synchronization Protocol (DPSP)**. This protocol breaks the traditional master-slave time synchronization paradigm that relies on a privileged reference frame (such as Earth), establishing a **Decentralized Trunk Anchoring Mechanism**. All nodes (whether Earth, Mars, or deep-space vehicles) act as equal branch observers, logically decoupling the "local time dilation factor (Γ)" from the "initial integration constant (t_0)": at the local end, at least two independent pulse units of different lengths are established. Each unit contains a local atomic clock timestamp of a quantum entanglement measurement event and a classical electromagnetic return pulse. Through multi-baseline cross-correlation operations and spacetime vector triangulation, the local branch curvature is single-point resolved. At the deep-space end, only an ultra-low-frequency (or single) initial handshake pulse is required to lock the initial integration constant t_0 . To cope with dynamically fluctuating time fields, the protocol introduces mechanisms such as

"adaptive pulse frequency scaling," "continuous curvature second-order compensation," and "causal backpressure buffering." Through continuous field integration along spacetime worldlines (Worldline Path Integration), each node independently reconstructs its causal order relative to the initial handshake ($t_{\text{trunk}} - t_0$) under dynamic gravity wells and non-inertial frames, achieving an engineering breakthrough from "intergalactic relative calibration" to "causal order reconstruction relative to initial handshake." Combined with astronomical calibration data from surface GPS (+38 $\mu\text{s/day}$) and Earth-Mars systems (47.9 $\mu\text{s/day}$), it is confirmed that under multi-frequency dispersion compensation, the protocol's theoretical resolution limit is bounded by current optical lattice clocks ($\mathcal{O}(10^{-16} \text{ s})$), while the engineering alignment error budget converges to the nanosecond scale ($\mathcal{O}(10^{-9} \text{ s})$). This framework naturally evolves into a "chronometric gravitational wave detection network," completing the full loop from theoretical causal deconstruction to deep-space spacetime engineering.

Introduction and Epistemological Disclaimer

This theory employs Information-theoretic Physics and topological geometry to structurally describe the causal chain of physical spacetime. Abstract concepts used in this paper—such as "static topological full graph," "restricted reading sequence," and "information propagation latency"—are strictly confined to the scope of information theory and mathematical modeling. They are intended to provide an objective deconstruction of physical observations without making any metaphysical assumptions or extensions regarding the ontology of the universe.

Chapter 1: Mathematical Boundary Failure and Domain Overflow of Special Relativity

Traditional Special Relativity faces two core mathematical breakdowns when dealing with extreme velocities, both originating from forcing the electromagnetic propagation limit (c) to be identical with the physical ontology of time:

1. The Negative Fallacy of Observed Time (t'):

In the transformation equation $t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$, when $v > c$, the numerator yields negative time.

Traditional interpretation directly claims this as "time reversal." In essence, this is a misinterpretation under geometric limit operations that mistakes "Wavefront Reversal" (receiving signals in reverse order) for the backward flow of physical time.

2. The Imaginary Fallacy of Proper Time (τ):

In the proper time formula $\tau = t\sqrt{1 - \frac{v^2}{c^2}}$, when $v > c$, an imaginary number i is calculated. This represents an "Undefined Domain" where the mathematical model transcends its scope of validity, rather than physical time turning into voidness.

Fundamental Qualitative Characterization of Boundary Failure

When $v > c$, the Lorentz transformation yields imaginary numbers and negative time, which historically has been directly interpreted as "physically impossible, hence superluminal motion does not exist." However, this interpretation confuses two distinct concepts:

- **Mathematical Model Failure:** The inputs to the equation exceed its valid range, thus producing invalid output values.
- **Physical Entity Impossibility:** Natural laws forbid the existence of the phenomenon based on first principles.

This paper points out that imaginary numbers and negative time under $v > c$ merely prove that the Lorentz transformation fails in that domain, and do not constitute a physical negation of superluminal propagation. This confusion is the logical root of the century-old time reversal paradoxes.

Chapter 2: Core Postulates and Global Causal Equation of the Tree-like Causal Latency Model (TCLM)

1. Core Postulates

To repair the mathematical breakdown of relativity under extreme conditions, we establish the following postulates to strictly decouple the "topological causal order (trunk)" from the "relative observational latency (branches)" of the physical world:

- **Postulate I: Topological Causal Order & Branch Curvature**
 - **Topological Causal Trunk (t_{trunk} / Topological Causal Order):** Represents the objective, irreversible causal sequence existing at the foundation of the universe (t_{trunk}). **It is hereby emphasized: t_{trunk} is NOT a Newtonian universal clock ticking uniformly across all space, NOR does it provide any preferred simultaneity slice.** t_{trunk} is a scalar topological causal parameter used solely to index the sequential settlement order of events (Directed Acyclic Graph, DAG). The existence and event settlement of all physical entities in the universe are equally rooted in this causal sequence, advancing strictly unidirectionally along the trunk.
 - **Geometric Relative Branches & Curvature:** Represents the relative observation trajectories and local proper time rates (τ) grown from the causal trunk for each physical entity. The essence of relative time flow rate is the curvature of the branch on a four-dimensional geometric manifold. Relativistic effects measured by local observers (such as time dilation and gravitational redshift) are the path-integrated results of geometric branches.
 - **Dynamic Environmental Adaptability (Continuous Curvature Field Floating):** Physical entities are not locked within the hierarchical structure of a specific celestial body. Spacetime curvature is essentially a continuous floating field that changes dynamically with local gravitational potential and velocity fields. As an object moves across galaxies, its physical branch adapts directly to the local continuous geometric curvature. All objects share the same global causal order while experiencing relative time differentials induced by instantaneous geometric curvature.
 - **Forward Growth Constraint:** Under the action of the standard topological causal operator ($\text{sgn}(\mathcal{C}) \equiv +1$), the growth vectors of all branches flow strictly unidirectionally downstream. The topological causal operator is strictly positive definite and never permits backward growth upstream toward the trunk within a single manifold. Historical coordinates belong to a topologically irreversible state, completely eliminating time reversal and paradox backdoors.
- **Postulate II: Decoupling of Event Ontology and Signal Carrier (Thunder and Lightning Isomorphism Model)**

- **Decoupling Event Ontology from Lightning:** Taking "the physical occurrence of thunder/lightning" and "the outward propagation of sound/light" as an isomorphic model: the instant a thunder event settles on the topological trunk (t_{trunk}) marks its settlement point on the topological causal order; the light received by an observer is merely the geometric branch signal propagating outward ($t_{\text{observed}} = t_{\text{trunk}} + \Delta t_{\text{latency}}$).
- **Distinction Between Propagation Limit and Time Ontology:** The light signal is merely an information carrier (geometric branch), and its propagation latency does not equal the occurrence time of the event itself. Even if one possesses superluminal means of propagation (perceiving the thunder event faster than light), it merely represents shortening the signal transmission latency (branch flattening), and never implies that the observer travels back prior to the occurrence (causal settlement) of the thunder event.

2. Global Causal Equation and Dual-Layer Geometric Decoupling

The TCLM global causal equation employs a dual-layer modular architecture that decouples macroscopic topological causal relationships from microscopic geometric field latency mechanisms:

$$t_{\text{observed}} = t_{\text{trunk}} + \Delta t_{\text{latency}}$$

where the microscopic geometric field operator (latency mechanism) is defined as:

$$\Delta t_{\text{latency}} \equiv \int_{\mathcal{P}} \frac{\sqrt{g_{ij}^{\text{branch}} dx^i dx^j}}{v_{\text{signal}}(x)}$$

(t_{trunk} is the causal order parameter satisfying strict monotonic increase; the integral term is determined by geometric distance, propagation speed, and branch curvature, remaining strictly positive in seconds).

Parameter Definitions:

- t_{observed} (Observed Time): The scalar time at which an observer receives information remnants in a specific coordinate system.
- t_{trunk} (Topological Causal Order Parameter): The underlying scalar index of event settlement, satisfying the strict monotonic increase postulate ($\frac{dt_{\text{trunk}}}{d\tau} > 0$). It represents the causal topological position of an event rather than a globally synchronized physical clock.
- $\Delta t_{\text{latency}}$ (Geometric Latency): The geometric delay accumulated as information propagates through branch space manifolds, measured in seconds (s), and strictly positive-definite ($\Delta t_{\text{latency}} \geq 0$).
- $\mathcal{P}$ (Propagation Trajectory): The physical geometric path traversed by the information carrier in the 3D branch spatial manifold.
- g_{ij}^{branch} (Branch Spatial Metric Tensor): A positive-definite geometric tensor describing local spatial curvature, gravitational potential, and propagation resistance limits ($i, j \in \{1, 2, 3\}$).
- dx^i, dx^j (Differential Spatial Displacements): Infinitesimal coordinate changes along the spatial propagation path.
- $v_{\text{signal}}(x)$ (Local Effective Propagation Velocity): The physical speed of information at coordinate x (e.g., effective light speed in local fields, satisfying $v_{\text{signal}}(x) > 0$), ensuring dimensional consistency $[\text{m}]/[\text{m/s}] = [\text{s}]$.

3. Historical Confusion Between Absolute and Relative Time: The TCLM Decoupling Perspective

In the history of physics, Newtonian absolute time and Einsteinian relative time have long been viewed as incompatible worldviews.

Newtonian absolute time assumes:

There exists a background time in the universe that flows uniformly and independently of matter, sharing a single universal "now."

Einsteinian relative time states:

Time is not an independent background, but forms a geometric manifold together with space; under different gravitational potentials and motion states, time flow rates naturally differ.

If viewed as mutually exclusive answers to the same question, conflict is inevitable.

However, TCLM points out that this conflict stems from mistaking descriptions of two different dimensions for the same phenomenon.

What Newton truly sought to preserve was not "identical ticking rates," but the "irreversibility of causal sequence."

What Einstein truly described was not "reversible causality," but that "local proper time rates vary with geometric conditions."

Within the TCLM framework:

- Newton's Absolute Time → Requirement for global causal order t_{trunk} .
- Einstein's Relative Time → Description of geometric branch latency.

The two do not contradict each other because they address fundamentally different dimensions. Newton saw the trunk of the tree; Einstein saw its branches. They simply stood before the same tree, pointing at different dimensions.

Thus, TCLM neither advocates a return to Newton nor negates Einstein. It merely restores "causality" and "time" to their proper, distinct places.

Chapter 3: Deep-Space Engineering Implementation: Dual-Pulse Synchronization Protocol (DPSP) and Initial Handshake Causal Order Reconstruction

Notation Definitions

- Γ_{local} : The local time dilation factor of any node relative to the initial handshake reference, $\Gamma \geq 1$.
- $\gamma_{B/A}$: The relative time transformation rate between two observation endpoints A and B , $\gamma_{B/A} = d\tau_B/dt_A$. These two parameters must not be confused.

0. Engineering Necessity and Paradigm Shift of DPSP

0.1 Engineering Motivation & Necessity

Traditional deep-space timekeeping architectures assume time flow rates to be predictable constants or static offsets. However, under TCLM, the local time dilation factor relative to the initial handshake (Γ) is intrinsically a

continuously floating field dynamically influenced by gravitational and velocity fields. This characteristic leads to three major engineering bottlenecks:

1. **Local Imperceptibility:**

When a single node is situated on a celestial surface (e.g., Earth's surface), local curvature gradient variations are extremely small, making time rate fluctuations virtually undetectable in short-term measurements. Observers easily mistake local time rate for a constant, ignoring dynamic corrections needed for alignment relative to the initial handshake.

2. **Interplanetary Scale Amplification:**

When spanning to deep-space endpoints (such as the Earth-Mars system), small differences in local curvature and potential continuously accumulate and amplify through time path integration. The Earth-Mars time rate differential reaches $\sim 47.9 \mu\text{s}/\text{day}$, dynamically fluctuating with orbital eccentricity, solar gravitational perturbations, and planetary interactions, rendering fixed-constant extrapolation invalid.

3. **Physical Infeasibility of Continuous Sync:**

Relying on continuous high-frequency two-way signaling faces two fundamental limits: **propagation delay limits at the speed of light** (e.g., Earth-Mars one-way delay ranges from 3 to 22 minutes, preventing real-time feedback); and **unpredictability of continuous dynamic micro-perturbations**, causing static mathematical extrapolations to diverge over time.

0.1.1 Applicability Boundary

DPSP is not intended to replace all deep-space timing architectures, nor is it necessary under all conditions. Its application criteria are:

1. **Sufficient Local Compute:** Nodes must possess computational power for cross-correlation, geometric reconstruction, and continuous integration. For low-power nodes, conventional telemetry synchronization remains sufficient.
2. **Interruption Tolerance:** If a mission allows continuous high-frequency handshakes over a stable link, conventional DSN methods suffice. DPSP's primary value lies in providing offline autonomous chronometry.
3. **Interplanetary or Farther Missions:** Near-Earth or short-duration missions derive minimal engineering benefit due to negligible relative time rate differentials.

In summary, DPSP is an offline causal reconstruction solution designed for "long-term, offline, high-compute" deep-space nodes, rather than a universal requirement.

0.2 Paradigm Shift: From "Earth Model Dependency" to "Local Physical Measurement"

The fundamental limitation of traditional deep-space timekeeping (e.g., Deep Space Network) lies not only in its master-slave ideology centered on Earth, but also in its complete reliance on Earth-calculated astronomical formulas and ephemeris models. Deep-space nodes lack the capacity to measure local time rate changes directly, passively receiving calibration parameters calculated by Earth. If communication breaks down or ephemeris models fail, nodes lose temporal autonomy.

Under TCLM, Earth time (t_{Earth}) and Mars time (t_{Mars}) are merely "geometric branch remnants" under local gravity wells and motion states; neither serves as an objective reference clock for the universe.

The breakthrough of DPSP lies in enabling deep-space nodes to perform "local physical measurements." Instead of querying Earth's gravitational field formulas, a node directly measures its own time flow rate change relative to the initial handshake reference (Γ_{local}) by comparing timestamps of local quantum entanglement events with classical return pulses. Earth-calculated models serve only as initial references, not as single sources of truth.

Conceptual Dimension	Traditional Deep-Space Timekeeping (DSN Type)	Dual-Pulse Synchronization Protocol (DPSP / TCLM)
Reference Frame	Earth Privileged Frame (Master-Slave)	Decentralized Unprivileged Nodes
Alignment Target	Track Earth Time t_{Earth}	Directly measure local time rate variation relative to initial handshake (Γ_{local})
Comm. Dependency	Requires frequent/continuous two-way handshakes	Requires only initial lock; subsequent propagation is computed autonomously locally
Physical Essence	Engineering compromise compensating signal delay	Observer's decoding of the universe's underlying topological causal order

Based on this paradigm shift, DPSP abandons continuous two-way phase-locking, **logically decoupling** the local time dilation factor (Γ) from the initial constant (t_0) to build a multi-layered dynamic control architecture.

While this paper uses the Earth-Mars system as a first-phase engineering calibration case, DPSP is not limited to interplanetary distances. As long as an initial handshake is executed once, subsequent autonomous reconstruction and communication blackout tolerance can logically extend to interstellar or intergalactic scales.

1. Layer 1: Local Spacetime Vector Triangulation (Earth-to-Earth / Mars-to-Mars)

Engineering Definition of "Dual-Pulse"

"Dual-pulse" in this protocol refers to establishing at least two baseline references of different lengths on the local node and simultaneously triggering two independent pulse units.

Each pulse unit comprises a local timestamp of a quantum entanglement measurement event and a local timestamp of a classical electromagnetic return.

A single baseline provides only a 1D projection of delay difference, incapable of uniquely resolving geometric tilt angles of the spacetime manifold.

Thus, the minimum requirement for dual-pulse triangulation is two baselines of different lengths; if a third baseline of a different length is available, closure validation can be performed to verify the uniqueness and stability of θ .

Operation of a Single Pulse Unit (Quantum-Classical Hybrid Measurement)

For any independent baseline (endpoints L_1, L_2), a single pulse measurement involves two inseparable physical steps, strictly obeying the Quantum No-Communication Theorem:

- Step 1: Quantum Entanglement Event Timestamping:**
Using pre-entangled quantum systems, measurements are triggered independently at L_1 and L_2 .

The single-sided reduced density matrix ($\rho = \frac{1}{2}I$) yields only random statistical eigenvalues, conveying no instantaneous information.

However, each measurement is a local event whose timestamp is recorded by the local atomic clock. This timestamp is unaffected by classical propagation path delays and is later cross-checked with the classical light return timestamp.

- **Step 2: Classical Light-Cone Return:**

Along the same baseline, the remote end (L_2) encodes its recorded data stream (quantum event timestamp + classical return timestamp) into a classical electromagnetic signal and transmits it back to L_1 .

Its propagation trajectory forms a standard local light cone, providing a geometric spatial delay reference.

Dual-Pulse Triangulation & Cross-Correlation Algorithm

The local system collects pulse data from two (or more) baseline references.

The local node (L_1) performs **cross-correlation** between its stored quantum event timestamps and incoming classical feature data streams:

$$R_{AB}(\tau) = \int A_{L_1}(t)B_{L_2}(t + \tau)dt$$

By comparing curve matching and local differential reconstruction across two or more baselines of different lengths, the system eliminates single-axis blind spots and precisely resolves the projection angle θ of the local clock's worldline within equivalent light-clock geometry (θ is not a Minkowski four-dimensional inner product angle).

The mapping between the local time dilation factor (relative to initial handshake) Γ_{local} and geometric tilt angle θ is defined as:

$$\cos \theta = \frac{d\tau}{d(t_{\text{trunk}} - t_0)} = \frac{1}{\Gamma_{\text{local}}(t)} \implies \Gamma_{\text{local}}(t) = \sec \theta \geq 1$$

Here $t_{\text{trunk}} - t_0$ is the causal order parameter relative to the initial handshake, not a directly observable global absolute time.

When local spacetime has no relative velocity or gravitational potential gradient, $\theta = 0^\circ$ and $\Gamma_{\text{local}} = 1$ (local flow rate equals trunk limit); under curvature tilt, $\Gamma_{\text{local}} \approx 1 + \frac{1}{2}\theta^2$.

This step **is conducted purely locally via multi-baseline data cross-check and geometric reconstruction**, requiring no real-time intergalactic communication.

When three or more baselines of distinct lengths are available, closure checks can validate the uniqueness and stability of θ .

In strong gravitational fields or extreme curvature environments, the exact formula $\Gamma_{\text{local}} = \sec \theta$ must be invoked; small-angle approximations are forbidden.

1.1 Geometric Insufficiency of a Single Baseline

In DPSP, using two or more baselines of different lengths is a mathematical necessity for angular resolution, not an engineering preference.

A single baseline provides only a scalar timestamp difference:

$$R_{AB}(\tau) = \int A_{L_1}(t)B_{L_2}(t + \tau)dt$$

This single scalar indicates that a delay difference exists, but cannot isolate its source:

- Time flow rate differential
- Spatial curvature differential
- Instrument response offset
- Random fluctuations

More fundamentally, a single scalar delay differential across one baseline length leaves the unknown angle θ mathematically degenerate with multiple valid solutions.

Any system attempting to resolve local geometric tilt using a single baseline suffers from rotational degeneracy in its solution space. Therefore, using two or more independent baselines of distinct lengths is physically necessary to break this degeneracy through differential delay analysis. Directional variation serves to mitigate anisotropic micro-perturbations or multipath interference, but length difference remains the primary requirement for resolving degeneracy.

1.2 Functional Positioning and Degeneracy Resolution of a Single Composite Pulse

Traditional single-direction classical light signals yield only a scalar delay, incapable of distinguishing spatial distance from time rate tilt.

In DPSP, a single pulse unit combines a quantum entanglement event timestamp and a classical light return timestamp:

1. **Quantum Entanglement Timestamp:** Recorded by local atomic clocks, unaffected by classical propagation latency.
2. **Classical Light Return Timestamp:** Recorded by local atomic clocks, subject to spatial distance and local curvature.

Comparing the two separates spatial propagation latency from time-dilation-induced latency. This functionality does not depend on specific interpretations of quantum mechanics, nor does it claim quantum entanglement directly observes the trunk.

Within TCLM, the quantum entanglement event timestamp acts as a second local reference with different noise/latency characteristics than classical light, rather than a direct readout of the trunk itself. This mapping is an interpretive correspondence within the framework, not a proven physical assertion.

2. Layer 2: Interplanetary Initial Handshake Pulse (Deep-Space Initial Constant Lock)

Because the global causal order reconstruction equation mathematically requires an unknown **initial integration constant** (t_0), local $\Gamma(t)$ variation analysis alone yields rate of progress, not the absolute offset between two time axes.

- **Execution Steps of a Single Handshake:**

1. A reference node (e.g., Earth) transmits a classical deep-space radio pulse containing an initial timestamp $t_{\text{Earth_start}}$ toward the target node (e.g., Mars).
2. The target node receives the pulse, calculates light propagation delay $\Delta t_{\text{propagation}}$ using orbital geometry and multi-frequency dispersion compensation, and locks the initial alignment constant t_0 between local time and reference time.

- **Autonomous Operation State:**

Once t_0 is locked, the target node can cease high-frequency deep-space communications, relying purely on real-time $\Gamma_{\text{Mars}}(\tau)$ from "Layer 1 local triangulation" to independently reconstruct its causal order relative to the initial handshake:

$$t_{\text{trunk_reconstructed}}(t) = t_0 + \int_{\tau_0}^{\tau} \frac{1}{\Gamma_{\text{Mars}}(\tau')} d\tau'$$

- **Blackout Re-synchronization Mechanism:**

If a deep-space node enters a solar conjunction blackout, its local continuous field integration engine independently tracks time rate changes without causal disruption.

The constant t_0 locked during the initial handshake is a one-time classical alignment parameter that does not drift due to local integration; re-executing a handshake after emerging from a blackout serves to re-confirm the alignment baseline and correct minor numerical drift accumulated during offline integration.

Geometric curvature measurements of time flow rates remain valid throughout the blackout.

3. Layer 3: Dynamic Control of Continuous Flow Rate (Adaptive Scaling, Secondary Compensation, and Backpressure)

To address frequency asymmetry and buffer overflows caused by continuous fluctuations in time flow rates ($\frac{d\Gamma}{dt} \neq 0$), the protocol incorporates the following dynamic control mechanisms:

- **Adaptive Pulse Frequency Scaling:**

When nodes exchange local time flow rate curves $\Gamma(t)$ during initial handshakes, a local node combines its own $\Gamma_{\text{source}}(t)$ with the target's $\Gamma_{\text{target}}(t)$ to derive the relative flow rate ratio $\lambda(t) = \frac{d\tau_{\text{target}}}{d\tau_{\text{source}}}$. The node in the higher time rate zone dynamically adjusts its pulse transmission interval:

$$\Delta t_{\text{source}} = \frac{\Delta \tau_{\text{target}}}{\lambda(t)}$$

If the target node resides in a lower time rate zone ($\lambda \ll 1$), the source node automatically lengthens its transmission interval to match the target's local processing frame rate, preventing buffer flooding.

- **Continuous Curvature & Acceleration Compensation:**

In dynamic gravitational or accelerating fields, dual-pulse headers append the node's instantaneous "dilation acceleration and gravitational potential derivative ($\frac{d\Gamma}{d\tau}$).". The receiving node uses a second-order Taylor expansion to fit continuous curvature:

$$\Gamma(\tau + \Delta\tau) \approx \Gamma(\tau) + \frac{d\Gamma}{d\tau} \Delta\tau + \frac{1}{2} \frac{d^2\Gamma}{d\tau^2} (\Delta\tau)^2$$

The receiving node predicts clock drift between discrete pulses locally, eliminating phase errors induced by continuous acceleration without requiring high-frequency signaling.

Phase perturbations caused by solar wind and interplanetary plasma can be treated as quasi-common-mode across nodes in the solar system. Multi-node cross-correlation partially suppresses these background perturbations without requiring explicit plasma modeling.

It should be noted that continuous curvature second-order compensation is fundamentally a discrete sampling system. When the gravitational field variation frequency approaches or exceeds the pulse sampling frequency, aliasing will occur, and the sampling will fail to resolve that variation. This is a physical limit of any discrete sampling system, not a design flaw of this protocol. In practice, higher sampling rates, adaptive triggering mechanisms, or prior gravitational field models must be used to mitigate this. The detailed engineering mapping is left to implementation teams.

- **Causal Backpressure and Queue Mechanism:**

If drastic shifts in flow rate ratios occur (e.g., a target node enters a deep gravity well) and local causal processing cannot keep pace with incoming data, the target node's ingress queue triggers a critical Backpressure Signal:

- The signal instructs upstream nodes to throttle data throughput.
- Overloaded event data is cached in the local node's "Causal Buffer Pool," queued strictly according to the monotonicity constraint $\frac{d\tau}{dt} > 0$.
- Information transmission is extended along latency channels. Provided buffer capacity is sufficient, this prevents queue overflow or event loss in low-time-rate branches.

This buffer pool is a local engineering queue bounded by hardware capacity and local causal processing rates. J^{cap} in theoretical models represents the information throughput limit of the discrete grid; its detailed engineering mapping is left to implementation teams.

4. Theoretical Closure and Engineering Boundary Decoupling

- **Theoretical Resolution Limit:** Excluding media and geometric perturbations, the theoretical resolution limit of this protocol under closed-loop geometry is bounded by optical lattice clock hardware at $\mathcal{O}(10^{-16} \text{ s})$. This represents a hardware limit, not a standalone protocol achievement.
- **Deep-Space Engineering Error Budget:** In practical deep-space environments, radio signals experience dispersion from solar wind plasma, gravitational lensing from the Sun, and ephemeris uncertainties (uncompensated uncertainties reach $\sim 10^{-6}$ to 10^{-8} s). It should be noted that a gap of approximately seven orders of magnitude exists between the local optical lattice clock hardware resolution ($\mathcal{O}(10^{-16} \text{ s})$) and the achievable deep-space link engineering alignment precision ($\mathcal{O}(10^{-9} \text{ s})$). This gap primarily arises from SNR attenuation in deep-space radio links, residual plasma dispersion, and ephemeris uncertainties, and constitutes the primary engineering challenge for physical deployment. This paper provides only the theoretical framework and error budget; detailed link engineering is left to implementation teams.

Chapter 4: Astrophysical Observational Data Calibration and Earth-Mars Time-Transfer Derivation

(Note: $\gamma_{C/A}$ in this chapter is defined as the **relative** time transformation rate between two endpoint observers, distinct from the local dilation factor Γ_{local} relative to initial handshake used in Chapter 3.)

1. Earth Calibration Case (GPS Satellites)

Considering combined Special and General Relativistic effects in Low Earth Orbit (velocity causes $-7 \mu\text{s}/\text{day}$, gravitational potential causes $+45 \mu\text{s}/\text{day}$), the net measured drift rate between Earth's surface and GPS branches is $+38 \mu\text{s}/\text{day}$. Substituting into the endpoint relative transformation formula:

$$\gamma_{\text{GPS}/\text{Earth}} = \frac{d\tau_{\text{GPS}}}{dt_{\text{Earth}}} \approx 1 + 4.40 \times 10^{-10}$$

This data serves as a baseline demonstration that single-point rate differentials within a localized gravity well are measurable and reconstructible.

2. Earth-Mars Interplanetary Alignment and Worldline Path Integration

Let Earth's surface (A) and Mars' surface (C) be independent physical branch endpoints. Based on mass and radius parameters, the static Earth-Mars relative rate differential is:

$$\gamma_{C/A} = \frac{d\tau_{\text{Mars}}}{dt_{\text{Earth}}} \approx 1 + \frac{GM_E}{R_E c^2} - \frac{GM_M}{R_M c^2} \approx 1 + 5.55 \times 10^{-10}$$

Thus, a Martian atomic clock runs faster than an Earth surface clock by approximately $47.9 \mu\text{s}/\text{day}$.

This model uses continuous field integration along spacetime worldlines as its primary dynamic alignment mechanism. All time flow rates are intrinsically continuous floating fields. Historical "local static chain transitivity rules (such as $\gamma_{C/A} = \gamma_{C/B} \cdot \gamma_{B/A}$)" are degenerate discrete cases when gravitational potential perturbations approach zero.

In relay communications, intermediate satellites function merely as "Edge Routers" for signal forwarding, not as essential nodes for physical causal computation. Mars requires only a single initial handshake pulse from Earth; combining Layer 3 dynamic frequency scaling and backpressure mechanisms with local triangulation and continuous integration over a 100 s window constrains engineering alignment errors to the nanosecond level. The ultimate limit is set by optical lattice clock hardware rather than the protocol itself. Actual integration window lengths depend on mission geometry and link conditions.

Chapter 5: Extension to Microscopic Effective Field Theory (EFT) and Falsifiability

This chapter presents a preliminary framework bridging TCLM with Quantum Field Theory and is not the core thesis of this paper. Microscopic correction terms, dispersion relations, and topological shear operators described here are initial concepts reserved for future research; their mathematical completeness and experimental validation fall outside the primary scope of this work.

Scope and EFT Boundary Statement

Microscopic operators, dispersion relations, and topological equations in this chapter are positioned as **phenomenological Effective Field Theory (EFT) models** under TCLM. At subluminal speeds ($v < c$) and macroscopic low-energy limits, this model naturally degenerates into standard Lorentz covariance in Special Relativity and metric forms in General Relativity. Based on the postulate that Planck length ($\ell_P \approx 1.6 \times 10^{-35} \text{ m}$) and Planck time ($t_P \approx 5.4 \times 10^{-44} \text{ s}$) represent absolute minimum physical units, TCLM acknowledges that spacetime possesses a discrete grid structure at extreme microscopic scales. Non-linear

terms in this chapter represent effective boundary responses of this microscopic discrete grid under macroscopic continuous manifold approximations.

1. Microscopic Dispersion Relation at the Planck Scale

At ultra-high frequencies and extreme energy states, discrete microscopic grid corrections are introduced into the dispersion relation:

$$E^2 = p^2 c^2 + m^2 c^4 \pm \xi \left(\frac{E}{E_{\text{Planck}}} \right)^n E^2$$

where the coefficient ξ and energy scale represent the physical pressure of "local branch curvature" on microscopic particles—i.e., effective damping by the discrete grid on high-energy wave packet propagation. If arrival time differences of high-energy Gamma-Ray Bursts (GRBs) deviate from standard relativistic predictions beyond a specific threshold, it serves as one of the falsifiable tests for non-linear latency effects of geometric branches at extreme energies.

The power exponent n remains an undetermined parameter to be constrained by observational limits on Lorentz invariance violation.

2. Quantum Entanglement Phase and Topological Latency Operator

The phase distortion equation for extreme perturbation latency ϵ_q within local density matrices is defined as:

$$\Delta\Phi_{\text{entangle}} = \frac{2\pi c}{\lambda_c} \left[\int_{\tau_1}^{\tau_2} \left(1 - \frac{1}{\Gamma(\tau)} \right) d\tau + \epsilon_q(\rho_{\text{vacuum}}) \right]$$

Using distributed local quantum clock networks for long-baseline classical pulse differential comparisons (an EFT conjecture test, not a DPSP engineering step), if the decay rate of EPR correlation functions matches predictions based on the reciprocal of local dilation factor $\Gamma(\tau)$, it serves as a testable conjecture within the EFT framework rather than part of the DPSP engineering loop.

3. Gravitational Field and Metric Tensor Coupling (Topological Information Cutoff and Vacuum Coupling)

Formalizing the causal trunk t_{trunk} as a background manifold, geometric branches manifest as 4D perturbation tensors:

$$g_{\mu\nu} = g_{\mu\nu}^{(0)} + h_{\mu\nu}^{\text{branch}}$$

(where $g_{ij}^{\text{branch}} \equiv g_{ij}^{(0)} + h_{ij}^{\text{branch}}$, $\mu, \nu \in \{0, 1, 2, 3\}$, $i, j \in \{1, 2, 3\}$).

Introducing the EFT Damping Divergence Operator:

$$\nabla_\mu h_{\text{branch}}^{\mu\nu} = \kappa \cdot \text{sgn}(\mathcal{C}) \cdot (J_{\text{info}}^\nu) \left[1 - \exp \left(-\frac{\|J_{\text{info}}^\nu\|}{J_{\text{cap}}^\nu} \right) \right] + Q_{\text{vacuum}}^\nu$$

where J^{cap} is the information throughput capacity limit absorbed by the spacetime grid via geometric area expansion (event horizon expansion); Q_{vacuum}^ν is the energy tensor dissipated into the microscopic background vacuum, ensuring global energy conservation $\nabla_\mu T_{\text{total}}^{\mu\nu} = 0$. This equation predicts that when ultra-strong field collisions exceed capacity limits, overloaded information flows induce information cutoff

damping and non-linear phase shifts on gravitational waves, offering a potential testing path via gravitational wave detectors (such as LIGO and LISA).

Chapter 6: Flagship Engineering Application: Chronometric Gravitational Wave Passive Detection

1. **Spacetime Perturbation Response:** Endpoint networks continuously executing DPSP curvature resolution (such as Earth-Mars clock networks) yield $\Gamma(t)$ curves that reflect not only orbital trajectories and potentials, but also transient distortions caused by passing gravitational waves (Metric Perturbation, $h_{\mu\nu}$).
2. **Chronometric Gravitational Wave Sensing:** This mechanism upgrades communication calibration networks into distributed "Chronometric Gravitational Wave Detection Networks." Through multi-node cross-correlation and differential operation of strontium optical lattice clocks, common-mode noise like solar wind can be partially suppressed, providing a potential observational path distinct from traditional detector arm-length constraints under specific conditions.

Chapter 7: Theoretical and Engineering Stress Testing & Boundary Defense (Q&A)

Question	Answer & Boundary Defense
Q1: How does TCLM explain the permanent age difference in the "Twin Paradox"?	The two clocks traversed different spatial trajectories (acceleration and gravity), accumulating different branch curvature delays. This represents geometric branch curvature differences, not backward travel on the trunk.
Q2: If superluminal propagation occurs, can signals be sent to the past or cause reverse causality?	According to TCLM's topological causal structure, no. Pure geometric acceleration causes only apparent reverse wavefront reception ($\Delta t_{\text{latency}} > 0$) without reversing causal vectors within a single manifold. The global causal operator is strictly positive ($\text{sgn}(\mathcal{C}) \equiv +1$), and historical coordinates are topologically irreversible, eliminating time travel and grandfather paradoxes.
Q3: Does time stop or reverse when matter falls into a black hole event horizon?	No. The event horizon is merely a geometric boundary where branch curvature reaches its limit. Matter's evolution along the causal trunk remains strictly forward-moving for observers.
Q4: Does TCLM's causal order violate tensor covariance?	Not at all. TCLM strictly separates "geometric observables" from "topological causal order." Local geometric observables (clock tick rate $d\tau$ and delay $\Delta t_{\text{latency}}$) strictly obey Lorentz covariance and General Relativistic metric equations (as local EFT); t_{trunk} acts solely as a causal ordering scalar for global DAGs, without interfering with metric tensor calculations in local inertial frames.
Q5: Does TCLM face relativistic mass/energy	Continuous superluminal motion (Type A) relies on local spatial metric manipulation to bypass geometric resistance; discrete superluminal motion

Question	Answer & Boundary Defense
divergence constraints near the speed of light?	(Type B) utilizes topological transitions without traversing continuous space, bypassing energy divergence limits.
Q6: Is TCLM a revival of Newtonian "Absolute Space" or "Aether"?	Absolutely not. Newtonian absolute spacetime assumes a universal clock ticking simultaneously at every point in rigid space. TCLM's t_{trunk} is not a clock , but a pure topological causal operator. TCLM provides no preferred simultaneity slices or mechanical drag reference frames. Measured clock time is always a path integral on branch manifolds.
Q7: Does superluminal collapse in quantum entanglement violate the No-Communication Theorem or disrupt the trunk?	No. Quantum entanglement serves only as a local measurement event within a pulse unit, recorded by local clocks and cross-checked later against classical returns. DPSP relies on multiple baselines rather than real-time quantum communication. Single-sided measurements yield random eigenvalues, transmitting no instantaneous data. Resolving geometric tilt locally via historical timestamps involves no superluminal information transfer, maintaining strict logical closure.
Q8: Will continuous time rate shifts or sudden traffic spikes crash DPSP?	Not under normal mission geometry and sampling density. The protocol incorporates adaptive pulse scaling, continuous curvature second-order compensation, and causal backpressure buffers to maintain data topological integrity and queue safety under acceleration and severe rate differentials. However, if the gravitational field variation frequency approaches or exceeds the pulse sampling frequency, aliasing will occur and the sampling will fail to resolve that variation. This is a physical limit of any discrete sampling system, and must be addressed with higher sampling rates or prior models.

Chapter 8: Conclusion

This paper deconstructs the mathematical overflow of Special Relativity under extreme speeds into "apparent reverse wavefront reception," establishing the complete architecture of the "topological causal trunk" and "geometric branch curvature" in TCLM. It proposes the "Dual-Pulse Synchronization Protocol (DPSP)" and its differential calculation method as engineering implementation standards.

The model identifies continuous worldline field integration as its primary dynamic core, supported by observational calibration data from GPS (+38 $\mu\text{s/day}$) and Earth-Mars systems (47.9 $\mu\text{s/day}$), supporting the feasibility of autonomous causal reconstruction relative to initial handshakes under dynamic gravity wells and non-inertial frames.

Regarding precision, this paper distinguishes two levels:

- Hardware Resolution Limit:** Current optical lattice clocks achieve local resolutions of $\mathcal{O}(10^{-16} \text{ s})$, representing a hardware limit rather than a standalone protocol achievement.
- Engineering Error Budget:** In deep-space environments, multi-frequency dispersion compensation and dynamic metric fitting constrain alignment errors relative to initial handshakes to the nanosecond level ($\mathcal{O}(10^{-9} \text{ s})$), satisfying deep-space navigation requirements.

Furthermore, the protocol can upgrade into a "Chronometric Gravitational Wave Detection Network," offering a potential observational approach distinct from traditional detector arm-length constraints under specific conditions.

The core contribution of this paper lies not in claiming a unified quantum gravity theory, but in demonstrating that imaginary numbers and negative time under extreme conditions represent domain overflows of mathematical models rather than physical time reversal. This epistemological correction offers a topological causal framework for understanding superluminal phenomena and deep-space timing engineering.

TCLM's Causal Tree provides a coexistent path for both Newtonian and Einsteinian concepts of time.

Appendix A: Speculative Extensions at Extreme Speeds

1. Prediction I: Apparent Time-axis Decoupling & Inversion

- **Observational Phenomenon:** When a superluminal entity (or wavefront) approaches an observer, the observer sees "the arrival image first, followed by photons from origin and midpoint."
- **Dual Mechanism:**
 - **Type A (Continuous Superluminal / Geometric Glide):** If an entity breaks light speed in continuous space, it creates a geometric analogue to supersonic sonic booms. Upon reaching closest geometric approach, the image appears and splits in two: one moves forward, while the other appears as a receding ghost image, showing superluminal motion is a reorganization of propagation delays.
 - **Type B (Discrete Superluminal / Topological Transition):** If achieved via topological jumps analogous to quantum tunneling, observers experience a "Discontinuous Frame": the image at point A remains while a new image instantly emerges at point B, potentially emitting spatial metric perturbation radiation.
- **Causal Defense:** This phenomenon is apparent wavefront reversal on geometric branches. On the topological trunk, event settlement remains strictly monotonic. **Under this model, observers can receive future images early via superluminal means, but cannot physically intervene in those events,** eliminating grandfather paradoxes.

2. Prediction II: Read-only Causal Horizon at Extreme Speeds

- As a carrier approaches and exceeds light speed, branch curvature enters an extreme high-pressure regime. Any attempt to transmit dynamic control commands ("Write") upstream to the trunk is rejected by topological operators.
- Under this speculative model, superluminal motion can serve as a passive observation carrier, but never as a tool to reverse cause and effect.